Speech Tech Week 8 AQ PA GA

1) Initial weights for a neural network are initialized using the following method **1 point**

- Weights are initialized randomly
- Weights are set to 0
- Weights are set to 1
- None of these

2) Consider a neural network which is trained on input x, the ground truth for this input is y, while the predicted output of the network is $\hat{y}$. Consider a loss function for training this network $\mathcal{L} = \frac{1}{2}\sum_{i=1}^{K}(y_i - \hat{y}_i)^2$. select appropriate name for this loss function. **1 point**

- Cross-Entropy Loss
- Mean Squared Error Loss
- Huber Loss
- None of these

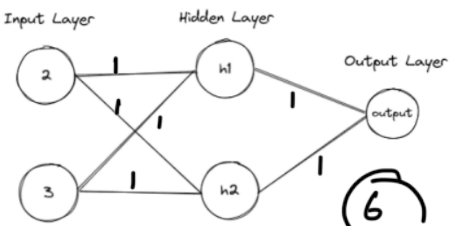
MSE

1) Using the gradient descent method to reduce the error function value, parameter (network weights) have to be changed in the direction of ____ slope. **1 point**

- Increasing
- Decreasing
- None of these

decreasing

2) For the neural network shown in figure, number of trainable weights in the network are **1 point**



6

- 6
- 8

1) State True or False: A local minima can work well for most practical deep-learning problems. **1 point**

- True
- False

1) State whether the following statement is true or false: Even though there are speaker variations, environment noise etc, for a given phoneme or sound, the mel filter bank features will be the same. **1 point**

diff.

- True
- False

2) State whether the following statement is true or false: We need alignments to train the neural network in HMM-DNN model. **1 point**

- True
- False

1) For every Mel-Filter bank feature, we try to align each feature with which type of following class? *1 point*

- [x] Phoneme
- [ ] Word
- [ ] Sentence
- [ ] syllable

phoneme 45x1

2) What is the fullform of CTC? *1 point*

- [x] Connectionist Temporal Classification
- [ ] Contextual Temporal Classification
- [ ] Con-Textual Classification
- [ ] Character Temporal Classification

1) State whether the following statement is true or false: CTC considers all possible paths such that the sequence order is maintained. *1 point*

- [x] True
- [ ] False

2) State whether the following statement is true or false: CTC attempts to maximize the sum of probability of all paths that ensure the correct sequence. *1 point*

- [x] True
- [ ] False

1) Consider a neural network which is trained on input x, the ground truth for this input is y, while the predicted output of the network is $\hat{y}$. Consider a loss function for training this network $\mathcal{L} = -\sum_{i=1}^{K} y_i log(\hat{y}_i)$. select appropriate name for this loss function. *1 point*

- [x] Cross-Entropy Loss
- [ ] Mean Squared Error Loss
- [ ] Huber Loss
- [ ] None of these

CEL

classification

2) In ________ machine learning task, softmax is applied at the output layer. *1 point*

- [ ] Regression
- [x] Classification
- [ ] None of these

3) Random initialization of the neural network can cause it to get stuck in an undesired local minima, we can adopt one of the the following method to partially prevent this from happening. *1 point*

- fine-tuning
- setting network weights to zero
- pre-training
- None of these

4) In a HMM-DNN, the number of neurons in the output layer will be equal to the number of ______ in the language

phemes

phonemes

1 point

5) State whether the following statement is true or false: We do not need supervision to train the neural network in HMM-DNN model. *1 point*

- True
- False

6) There is high possibility that phonemes are going to changed after every 25ms window. *1 point*

- True
- False

7) State whether the following statement is true or false: CTC allows for end-to-end speech recognition. *1 point*

- True
- False

1) Let vector $z = [3, 2, 1]^T$ be the input to a softmax layer, select the correct vector output of the softmax layer. *1 point*

- $[0.3, 0.2, 0.1]^T$
- $[0.7, 0.1, 0.2]^T$
- $[0, 0, 0]^T$
- $[0.66, 0.24, 0.1]^T$

$$\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\frac{e^3}{e^3+e^2+e^1} = \frac{20.08}{20.08+7.38+2.71} = \frac{20.08}{30.17} = 0.66$$

Yes, the answer is correct.
Score: 1

Accepted Answers:

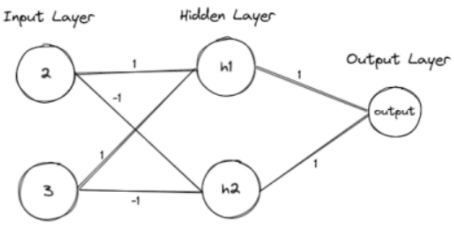
$[0.66, 0.24, 0.1]^T$

2) Let vector z be the output of a softmax layer, Select appropriate choice for $\sum_{i=1}^{K} z_i$. *1 point*

1

- -1
- 1
- 0
- None of these

3) For the neural network shown in figure calculate the output at the final layer. **_1 point_**

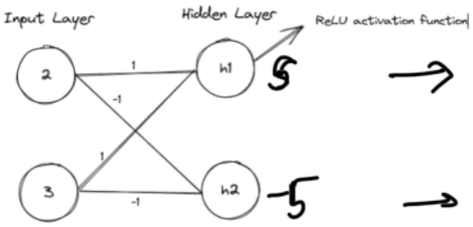


- -1
- 1
- 0
- None of these

$h_1 = 2 \times 1 + 3 \times 1 = 5$ $O = 5 \times 1 + -1 \times 5$

$h_2 = 2 \times -1 + 3 \times -1 = -5$ $= 0$

4) For the neural network shown in figure, $h = [h1, h2]^T$ represents the output of the hidden layer. The layer has a ReLU activation function, determine the output of the hidden layer **_1 point_**
hint: $h = f_{activation}(Wx)$).



$5 \rightarrow 5$

$-5 \rightarrow 0$

$[5, 0]^T$

- $[0, 5]^T$
- $[5, 0]^T$
- $[5, -5]^T$
- $[-5, 0]^T$

5) In a HMM-DNN, which of the following probabilities in a regular HMM-GMM model is replaced by probability from HMM-DNN? **_1 point_**

- Transition probability
- Observation probability
- Word probability
- None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:

Observation probability

6) Forced alignment is better than CTC alignment. **_1 point_**

- True
- False

7) CTC alignment can also be used for cursive hand writing recognition. *1 point*

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:

True

8) Which of the following ideas is/are exploited by CTC? *1 point*

- Frame level alignments are very important
- The sequence of phones is important
- both (a) and (b)
- None of the above